

SHELIKHOVA, Russian Federation

WMO: 259220

Lat: 60.717N

Lon: 155.883E

Elev: 109

StdP: 14.64

Time Zone: 11.00 (E11)

Period: 88-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.7	-27.8	1.2	-15.7	-24.9	1.4	-12.8	45.5	16.1	39.0	19.6	12.7	290	0.897

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	64.5	56.6	61.8	55.4	59.7	54.9	58.4	61.7	57.2	59.9	56.1	58.7	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.0	69.6	59.5	55.8	66.5	58.3	54.7	63.9	57.3	25.4	61.9	24.6	60.1	23.9	58.9	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.6	34.5	29.4	-20.3	73.1	3.9	4.8	-23.1	76.6	-25.4	79.4	-27.6	82.1	-30.5	85.6		
			-20.4	60.5	3.4	1.3	-22.9	61.4	-24.9	62.2	-26.8	62.9	-29.3	63.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	4.4	4.7	11.0	23.8	35.8	47.7	54.7	54.1	45.9	31.6	17.7	9.3
	DBStd	19.92	11.17	10.72	10.38	7.06	5.24	5.01	3.66	3.35	5.10	6.79	9.46	9.76
	HDD50	8169	1414	1268	1208	787	441	102	2	7	141	570	969	1260
	HDD65	13309	1879	1688	1673	1237	905	519	319	337	573	1035	1419	1725
	CDD50	335	0	0	0	0	1	33	149	135	17	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	5	0	0	0	0	0	0	5	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	12.7	15.0	14.1	14.3	12.1	9.5	8.8	8.8	9.6	11.1	14.1	17.7	17.9
(15) Precipitation	PrecAvg	18.4	0.6	0.6	1.0	0.9	0.8	1.1	1.8	3.6	2.8	2.6	1.7	0.7
	PrecMax	25.2	1.3	1.5	2.6	4.8	2.5	2.9	4.6	10.0	7.4	5.8	6.3	2.6
	PrecMin	11.9	0.0	0.0	0.1	0.1	0.1	0.0	0.1	0.4	0.4	0.5	0.0	0.1
	PrecStd	4.1	0.4	0.4	0.7	1.0	0.6	0.9	1.3	2.1	2.0	1.3	1.4	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	33.4	30.5	40.1	56.5	68.1	71.7	65.5	60.2	48.3	38.4	32.0
		MCWB	31.2	32.1	27.5	34.1	46.1	54.5	59.0	57.0	52.8	44.2	36.5	31.0
	2%	DB	28.1	29.1	28.6	36.6	49.4	60.6	66.5	63.0	56.4	46.5	36.3	28.9
		MCWB	27.2	28.1	26.9	32.7	43.0	52.1	57.5	56.1	51.4	43.1	34.9	27.3
	5%	DB	24.6	24.1	27.1	33.7	45.8	57.6	62.9	61.0	54.5	44.1	33.9	25.9
		MCWB	23.3	22.6	25.8	30.1	40.5	51.1	56.6	55.6	50.3	40.2	32.9	24.3
	10%	DB	20.5	20.3	24.8	32.1	43.4	55.4	60.3	59.4	52.9	41.2	30.9	23.0
		MCWB	18.8	18.7	23.0	30.0	38.9	50.4	55.4	55.1	49.2	37.1	29.7	21.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.3	32.3	28.6	34.7	46.3	56.4	60.9	60.1	55.8	46.0	36.8	31.3
		MCDB	31.4	33.3	29.3	38.2	53.6	62.8	67.1	62.4	57.5	47.5	38.1	32.2
	2%	WB	27.5	28.4	27.5	33.1	43.4	53.8	59.0	58.2	53.6	44.1	35.1	27.5
		MCDB	28.1	29.2	28.4	35.8	48.5	57.8	63.6	60.2	55.0	45.9	36.3	28.5
	5%	WB	23.4	23.1	26.0	31.4	41.4	52.4	57.6	57.2	51.7	41.0	33.0	24.4
		MCDB	24.5	23.8	27.1	32.8	45.3	56.1	61.2	59.2	53.2	43.3	33.8	25.7
	10%	WB	18.7	18.8	23.4	30.3	39.5	51.1	56.4	56.2	50.2	37.5	29.9	21.5
		MCDB	20.4	20.4	24.5	31.6	42.6	54.8	59.4	58.4	52.2	40.4	30.6	22.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.0	9.2	8.3	7.7	8.0	6.7	7.4	8.6	9.2	8.0	7.5
		MCDBR	8.5	6.5	7.0	7.2	11.8	12.7	13.2	10.1	9.2	9.3	6.3	9.3
		MCWBR	8.4	6.4	6.2	5.7	7.7	8.4	7.9	6.8	6.8	7.6	6.2	9.0
	5% WB	MCDBR	8.0	7.1	5.7	6.4	10.9	10.6	9.4	7.9	6.2	8.3	6.4	9.1
		MCWBR	8.2	7.5	5.6	5.2	7.4	7.7	6.4	6.1	5.5	7.8	6.2	8.8
(40) Clear-Sky Solar Irradiance	taub	0.219	0.265	0.321	0.393	0.423	0.401	0.455	0.425	0.336	0.309	0.253	0.208	
	taud	2.047	2.091	1.978	1.853	1.988	2.290	2.178	2.249	2.481	2.277	2.099	1.976	
	Ebn at Noon	169	226	246	247	251	262	242	240	250	209	148	116	
	Edh at Noon	13	24	38	52	49	37	41	36	23	20	12	9	
(44) All-Sky Solar Radiation	RadAvg	123	414	972	1563	1700	1677	1575	1165	796	405	139	54	
	RadStd	14	29	61	109	158	173	130	136	107	49	14	5	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page